ECA1803 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO1 Engineering Design Assignment: Standards-Compliant Optical Fiber Link Design

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Summary of Given Link Parameters and Operational Constraints

Parameter	Given Value	Parameter	Given Value
Transmission Reach (L)	60 km	Operating Wavelength (λ)	1550 nm (C-band)
Candidate Fiber A	SI: n1=1.4500, n2=1.4450, a=4.0 μm	Candidate Fiber B	SI: n1=1.480, n2=1.465, a=25 μm
Fiber Attenuation (α)	Abs: 0.04, Ray: 0.11, Bend: 0.03 dB/km	Transmitter Power (P_tx)	+3.0 dBm (DFB laser)
Connectors & Splices	4 Conn @ 0.4 dB, 10 Splices @ 0.08 dB	Rx Sensitivity & Margin	P_sens = -20 dBm, Margin = 6.0 dB
Chromatic Dispersion	G.652.D: 17 ps/(nm·km); G.655: 4.5	DCF Specifications	D_DCF = -100 ps/(nm·km), α = 0.5 dB/km
Source Spectral Width (Δλ)	0.4 nm; Input Pulse Width = 30 ps	OSNR Study & EDFA	P_sig = -16 dBm, P_ase = -39 dBm; Gain=14 dB

Stage 1 – Interpret the Communication Requirement

To design a robust, standards-compliant 60 km optical link capable of transporting 10 Gb/s traffic at 1550 nm with forward upgradeability to 100 Gb/s, an engineering backward-chaining methodology is applied, working backward from receiver requirements to the optical source.

1. Purpose of 1550 nm Operating Condition: The link operates at 1550 nm within the optical telecommunication C-band (1530–1565 nm), representing the absolute global attenuation minimum of fused silica glass (total attenuation ~0.18 dB/km). Rayleigh scattering, which scales inversely with the fourth power of wavelength ($1/\lambda^4$), is reduced by over 60% compared to 1310 nm. Furthermore, 1550 nm aligns directly with the stimulated emission band of Erbium-Doped Fiber Amplifiers (EDFAs), enabling low-noise in-line optical amplification.

2. Ray vs. Wave Optics, Coherence, and Polarization: Geometric ray optics (Snell's law and total internal reflection) is valid only when core dimensions are much larger than wavelength ($2a \gg \lambda$), as in Candidate Fiber B ($2a = 50\ \mu m \approx 32\lambda$). In Candidate Fiber A ($2a = 8.0\ \mu m \approx 5.16\lambda$), wave optics based on Maxwell's equations is required to model modal field distribution, evanescent cladding decay, and modal cut-off. With source spectral width $\Delta\lambda = 0.4$ nm (optical bandwidth ~50 GHz), the coherence length is $L_c \approx 6.0$ mm, providing sufficient temporal coherence for direct-detection NRZ transmission without causing destructive multi-path speckle noise. Single-mode fibers support two degenerate orthogonal polarization modes (HE11_x, HE11_y); fiber core asymmetries break degeneracy, creating modal birefringence and Polarization Mode Dispersion (PMD).

3. Light-Acceptance and Guided Modes: Numerical aperture ($NA = \sqrt{(n1^2 - n2^2)}$) defines light-gathering capacity from the source. The normalized frequency parameter $V = (2\pi a/\lambda)NA$ dictates modal behavior; single-mode operation requires $V \leq 2.4048$ (LP11 cut-off). Candidate Fiber A must be evaluated to ensure strictly fundamental LP01 propagation.

4. ITU-T Fiber Categories for 60 km Reach: ITU-T G.652.D (standard single-mode fiber, SSMF) offers low attenuation and zero water-peak with chromatic dispersion $D \approx 17$ ps/(nm·km) at 1550 nm. ITU-T G.655 (non-zero dispersion-shifted fiber) provides $D \approx 4.5$ ps/(nm·km) to suppress non-linearities in DWDM. At the building entry, tight conduit routing requires ITU-T G.657.A bend-insensitive fiber, which tolerates bend radii down to 10 mm without excessive leakage.

5. Loss Mechanisms and Power Budget Quantities: Power loss occurs through material absorption (0.04 dB/km), Rayleigh scattering (0.11 dB/km), distributed microbending (0.03 dB/km), discrete connector losses (4 units @ 0.4 dB), fusion splices (10 units @ 0.08 dB), and localized building-entry bends. The received power must exceed the -20.0 dBm sensitivity while preserving a mandatory 6.0 dB system margin.

6. Dispersion Quantities and Pulse Integrity: Modal dispersion dominates multimode fibers. Single-mode fibers suffer from chromatic dispersion ($\Delta\tau = D \cdot \Delta\lambda \cdot L$) and PMD ($0.1\sqrt{L}$). At 10 Gb/s (bit period Tb = 100 ps), pulse spreading must be strictly controlled (typically < 0.35 Tb) using Dispersion-Compensating Fiber (DCF) to prevent inter-symbol interference (ISI).

7. Signal Quality (OSNR) and Relation to BER: Amplified Spontaneous Emission (ASE) noise from in-line EDFAs degrades the Optical Signal-to-Noise Ratio (OSNR) measured across a 0.1 nm reference bandwidth. OSNR directly dictates electrical signal-to-noise ratio, Q-factor, and Bit Error Rate ($BER = 0.5\ \mathrm{erfc}(Q/\sqrt{2})$). A minimum OSNR of 17.0 dB is required to guarantee uncorrected $BER < 10^{-12}$.

> **STAGE 1 DELIVERABLE – CONCISE DESIGN REQUIREMENT STATEMENT**
> The proposed 60 km optical fiber link must reliably transmit 10 Gb/s data at 1550 nm (with architectural provision for 100 Gb/s upgrade) by utilizing single-mode guided wave propagation ($V \leq 2.405$) to eliminate modal dispersion, suppressing building-entry bend loss to ≤ 0.10 dB using ITU-T G.657.A fiber, maintaining an operational power margin of ≥ 6.0 dB above the -20 dBm receiver sensitivity, mitigating chromatic pulse broadening via dispersion compensation to eliminate ISI, and ensuring an end-of-life OSNR ≥ 17.0 dB with link PMD < 1.0 ps.

Stage 2 – Investigate & Select Fiber / Propagation Configuration

To select the proper fiber, the light-acceptance capability (NA), normalized frequency (V-number), guided mode count, and cut-off conditions of Candidate Fibers A and B are quantitatively evaluated.

1. Numerical Aperture (NA) Evaluation: Numerical Aperture evaluates light gathering from the optical source: $NA = \sqrt{(n1^2 - n2^2)}$.

• Candidate Fiber A: $NA_A = \sqrt{(1.4500^2 - 1.4450^2)} = \sqrt{(2.102500 - 2.088025)} = \sqrt{0.014475} = 0.1203$

• Candidate Fiber B: $NA_B = \sqrt{(1.4800^2 - 1.4650^2)} = \sqrt{(2.190400 - 2.146225)} = \sqrt{0.044175} = 0.2102$

Fiber B provides a wider acceptance cone ($\theta_a = 12.13°$ vs 6.91°), gathering more power from incoherent sources. However, this larger core allows hundreds of guided modes, inducing fatal modal dispersion over long spans.

2. Normalized Frequency (V-Number) at 1550 nm ($\lambda = 1.55$ μm): The normalized frequency parameter combines wavelength, core size, and index difference: $V = (2\pi a / \lambda) \cdot NA$.

• Candidate Fiber A (a = 4.0 μm): $V_A = [2\pi \cdot 4.0\ \mu m \cdot 0.1203] / 1.55\ \mu m = 3.0238 / 1.55 = 1.951$

• Candidate Fiber B (a = 25.0 μm): $V_B = [2\pi \cdot 25.0\ \mu m \cdot 0.2102] / 1.55\ \mu m = 33.0177 / 1.55 = 21.30$

Because $V_A = 1.951 < 2.4048$, Fiber A operates strictly in the fundamental single-mode regime (LP01). In contrast, $V_B = 21.30 \gg 2.4048$ places Fiber B deep into the multimode regime.

3. Mode Count Approximation & Cut-Off Wavelength: For step-index fibers, total guided modes (both polarizations) is $M_SI \approx V^2 / 2$. For graded-index fibers, $M_GI \approx V^2 / 4$.

• Fiber B Mode Count: $M_B \approx (21.30)^2 / 2 = 453.69 / 2 \approx 227$ guided modes.

• Fiber A Cut-off Wavelength (λ_c): Single-mode cut-off occurs when V = 2.4048:

$\lambda_c = (2\pi \cdot a \cdot NA) / 2.4048 = (2\pi \cdot 4.0\ \mu m \cdot 0.1203) / 2.4048 = 3.0238 / 2.4048 = 1.257\ \mu m = 1257$ nm.

Because operating wavelength $\lambda = 1550$ nm > $\lambda_c = 1257$ nm, Fiber A is strictly single-mode.

4. Ray Theory vs. Wave Optics Validity: Ray theory assumes light travels as rectilinear geometric rays obeying Snell's law and total internal reflection. This macroscopic model is valid only when core dimensions are much larger than wavelength ($a \gg \lambda$, $V \gg 1$), which applies to Fiber B (a = 25 μm = 16.1λ, V = 21.30). In Fiber A (a = 4.0 μm = 2.58λ, V = 1.951), core dimensions approach the optical wavelength; electromagnetic wave boundary conditions, transverse modal interference, and evanescent cladding penetration dominate. Therefore, wave optics is strictly required for Fiber A.

5. Fiber Alternatives & Bandwidth-Distance Comparison: Comparison of candidate fiber propagation alternatives:

Propagation Parameter	Candidate Fiber B (SI-MMF)	Graded-Index MMF (GI-MMF)	Candidate Fiber A (SMF - G.652.D)
Modal Delay Spread (Δt)	$\Delta t \approx L \cdot n1 \cdot \Delta / c \approx 3000$ ns	$\Delta t \approx L \cdot n1 \cdot \Delta^2 / (8c) \approx 3.8$ ns	**0.00 ns (Completely Absent)**
Bandwidth-Distance Product	20 – 50 MHz·km	1 – 2 GHz·km	**> 100 GHz·km (Dispersion Limited)**
Maximum Reach @ 10 Gb/s	< 15 meters (Catastrophic ISI)	< 500 meters	**> 60 km (With Compensation)**
ITU-T Standards Mapping	Legacy LAN (OM1/OM2)	Building Risers (OM3/OM4)	**Telecom Core/Metro (ITU-T G.652.D)**

> **DESIGN DECISION 1 – CREATE THE PROPOSED FIBER CONFIGURATION**
> I recommend ITU-T G.652.D single-mode fiber propagation/fiber configuration because it strictly operates in the single-mode regime ($V = 1.951 < 2.4048$) at 1550 nm with single-mode cut-off at 1257 nm, completely eliminating the 3000 ns intermodal delay spread of multimode Fiber B, providing lowest optical attenuation (0.18 dB/km), and delivering the high bandwidth-distance capability (>100 GHz·km) required to transport 10 Gb/s and support 100 Gb/s over 60 km.

Stage 3 – Develop and Evaluate the Optical Power Budget

Working backward from receiver sensitivity (P_{sens} = -20.0 dBm) and required system margin (6.0 dB), the minimum received power required is $P_{rx,req}$ = -20.0 + 6.0 = -14.0 dBm.

1. Fiber Attenuation Coefficient: The total attenuation coefficient is the sum of all distributed loss mechanisms:

$\alpha_{total} = \alpha_{absorption} + \alpha_{Rayleigh} + \alpha_{bending} = 0.04 + 0.11 + 0.03 = 0.18$ dB/km.

2. Main Span Loss: Total fiber attenuation across the 60 km span: $A_{fiber} = 60 \text{ km} \times 0.18 \text{ dB/km} = 10.80$ dB.

3. Hardware Insertion Losses: Passive component losses:

• Connectors (4 units @ 0.40 dB): $A_{conn} = 4 \times 0.40 \text{ dB} = 1.60$ dB.

• Fusion Splices (10 units @ 0.08 dB): $A_{splice} = 10 \times 0.08 \text{ dB} = 0.80$ dB.

4. Building-Entry Bend Loss & Link Budget Comparison: Building-entry bend loss and link budget comparison:

• Baseline Case (G.652.D throughout): Entry bend loss = 0.80 dB.

Total Loss = 10.80 + 1.60 + 0.80 + 0.80 = 14.00 dB.

Received Power: $P_{rx} = P_{tx} - A_{total} = +3.0 \text{ dBm} - 14.00 \text{ dB} = -11.00$ dBm.

Available Margin: M = -11.00 dBm - (-20.00 dBm) = 9.00 dB (3.0 dB above required 6 dB margin).

• Optimized Case (G.657.A at building entry): Entry bend loss = 0.10 dB (saves 0.70 dB!).

Total Loss = 10.80 + 1.60 + 0.80 + 0.10 = 13.30 dB.

Received Power: $P_{rx} = P_{tx} - A_{total} = +3.0 \text{ dBm} - 13.30 \text{ dB} = -10.30$ dBm.

Available Margin: M = -10.30 dBm - (-20.00 dBm) = 9.70 dB (3.7 dB above required 6 dB margin).

Power Budget Component	G.652.D Baseline	G.657.A Optimized	Design Specification & Remarks
Transmitter Launch Power (P_{tx})	+3.00 dBm	+3.00 dBm	1550 nm DFB Laser Source
60 km Fiber Cable Attenuation	10.80 dB	10.80 dB	α_{total} = 0.18 dB/km × 60 km
Patch Panel Connectors (4 units)	1.60 dB	1.60 dB	4 × 0.40 dB/connector
Field Fusion Splices (10 units)	0.80 dB	0.80 dB	10 × 0.08 dB/splice
Building-Entry Bend Loss	0.80 dB	0.10 dB	G.657.A saves 0.70 dB tight bend penalty
Total Passive Link Loss (A_{total})	14.00 dB	13.30 dB	Passive link attenuation
Received Optical Power (P_{rx})	-11.00 dBm	-10.30 dBm	Operating Margin: 9.70 dB (vs 6.0 dB req.)

5. Dominant Loss Mechanism & Practical Loss Reductions: Rayleigh scattering (0.11 dB/km) is the dominant loss mechanism, contributing 61.1% of total fiber attenuation. It arises from fundamental microscopic thermodynamic density fluctuations in silica glass and cannot be altered in deployed glass.

• Practical Action 1 (Building-Entry Optimization): Deploy ITU-T G.657.A bend-insensitive fiber for the building drop conduit, reducing bend loss from 0.80 dB to 0.10 dB.

• Practical Action 2 (Splicing & Connector Control): Use core-alignment fusion splicers (<0.04 dB/splice) and angled physical contact connectors (LC/APC, <0.25 dB) with rigorous inspection.

> **DESIGN DECISION 2 – CREATE THE PROPOSED POWER-BUDGET CONFIGURATION**
> I recommend **standard single-mode fiber with ITU-T G.657.A bend-insensitive fiber at the building entry** power-budget configuration because **it achieves a total passive link attenuation of 13.30 dB and a received optical power of -10.30 dBm, providing a robust 9.70 dB operating margin that exceeds the mandatory 6.0 dB requirement by 3.70 dB, while effectively eliminating the 0.80 dB tight-bend penalty at the building conduit entry.**

Stage 4 – Investigate & Select Dispersion / PMD Configuration

1. Modal Dispersion Principles: Single-mode fiber eliminates intermodal dispersion because optical energy travels in a single spatial mode (LP01). In step-index multimode fiber, different guided modes travel along distinct geometric paths with differing propagation angles, producing extreme intermodal delay spread ($\Delta t \approx 3000$ ns over 60 km). Graded-index fiber parabolic profiling equalizes transit times (higher-order modes travel faster in lower-index cladding regions), reducing delay spread to ~3.8 ns; however, this remains unacceptable for 10 Gb/s (Tb = 100 ps).

2. Chromatic Dispersion Broadening: Chromatic dispersion broadening is given by $\Delta\tau = D \cdot \Delta\lambda \cdot L$, where $\Delta\lambda = 0.4$ nm and L = 60 km.

• ITU-T G.652.D (D = 17 ps/(nm·km)): $\Delta\tau = 17 \times 0.4 \times 60 = 408.0$ ps.

• ITU-T G.655 (D = 4.5 ps/(nm·km)): $\Delta\tau = 4.5 \times 0.4 \times 60 = 108.0$ ps.

3. Pulse Broadening and Inter-Symbol Interference (ISI): Combining input pulse width T0 = 30 ps with dispersion broadening via Root-Sum-Square (RSS): $T_{out} = \sqrt{(T0^2 + \Delta\tau^2)}$.

• G.652.D: $T_{out} = \sqrt{(30^2 + 408^2)} = \sqrt{(900 + 166464)} = \sqrt{167364} = 409.1$ ps.

Ratio to bit period: $T_{out} / Tb = 409.1 / 100 = 4.09$ bit intervals! Optical pulses spread across four adjacent bit slots, causing complete eye closure and catastrophic ISI. Transmission is impossible without dispersion compensation.

• G.655: $T_{out} = \sqrt{(30^2 + 108^2)} = 112.1$ ps > Tb = 100 ps. Even with lower dispersion, pulses spill across bit boundaries, causing excessive eye-closure penalty.

4. Dispersion-Compensating Fiber (DCF) Calculation: Complete first-order compensation requires $D_{SMF} \cdot L_{SMF} + D_{DCF} \cdot L_{DCF} = 0$.

• Accumulated dispersion in 60 km G.652.D: $D_{accum} = +17$ ps/(nm·km) × 60 km = +1020 ps/nm.

• Required DCF Length ($D_{DCF} = -100$ ps/(nm·km)): $L_{DCF} = 1020 / 100 = 10.20$ km.

• Added DCF Insertion Loss: $A_{DCF} = 10.20$ km × 0.50 dB/km = 5.10 dB.

• Revised Total Passive Link Loss: $A_{total,rev} = 13.30$ dB + 5.10 dB = 18.40 dB.

• Revised Received Power: $P_{rx,rev} = +3.00$ dBm - 18.40 dB = -15.40 dBm.

• Resulting Passive Margin: $M_{rev} = -15.40$ dBm - (-20.00 dBm) = 4.60 dB.

Crucial Observation: The revised passive margin of 4.60 dB falls below the required 6.0 dB system margin. This proves that deploying DCF mandates optical amplification to restore the power margin.

5. Polarization Mode Dispersion (PMD) & Birefringence: Link PMD accumulates stochastically: $\Delta\tau_PMD = D_PMD \cdot \sqrt{L} = 0.1$ ps/√km × √60 km = 0.775 ps.

• 10 Gb/s Limit: $PMD_limit = 0.1 \cdot Tb = 0.1 \times 100$ ps = 10.0 ps. Since 0.775 ps ≪ 10.0 ps, PMD is fully acceptable.

• 100 Gb/s Upgrade (Tb ≈ 10 ps): $PMD_limit = 0.1 \times 10$ ps = 1.0 ps. Since 0.775 ps < 1.0 ps, the fiber physical plant supports 100 Gb/s.

• PMD Mitigation Technique: Adaptive polarization-diversity electronic dispersion compensation (EDC) in coherent DSP receivers.

• Beat Length ($L_B = \lambda / B$ at $\lambda = 1.55$ μm):

- Standard Fiber ($B = 1 \times 10^{-7}$): $L_B = (1.55 \times 10^{-6}) / (1 \times 10^{-7}) = 15.5$ m.
- PM Fiber ($B = 4 \times 10^{-4}$): $L_B = (1.55 \times 10^{-6}) / (4 \times 10^{-4}) = 3.875$ mm = 0.00388 m.

• Role of PM Fiber: PM fiber maintains high internal birefringence to preserve linear polarization states. It is unnecessary and cost-prohibitive for 60 km transmission lines; standard single-mode fiber with polarization-insensitive receivers is the telecom standard.

> **DESIGN DECISION 3 – CREATE THE DISPERSION / PMD-MANAGEMENT CONFIGURATION**
> I recommend **ITU-T G.652.D fiber paired with a 10.2 km lumped DCF module** dispersion/PMD configuration because **uncompensated chromatic dispersion (408 ps) expands pulses over 4 bit slots causing catastrophic ISI at 10 Gb/s. The 10.2 km DCF completely cancels the +1020 ps/nm accumulated dispersion (restoring pulse width to 30 ps), while the fiber's PMD of 0.775 ps easily meets the 10 ps limit for 10 Gb/s and the 1.0 ps limit for 100 Gb/s.**

Stage 5 – Evaluate OSNR and the Amplification Strategy

1. Performance Metric & Reference Bandwidth: Optical Signal-to-Noise Ratio (OSNR) measured within a standardized optical reference bandwidth of 0.1 nm (12.5 GHz at 1550 nm) is the primary figure of merit for optical signal quality against Amplified Spontaneous Emission (ASE) noise.

2. Initial OSNR Evaluation: Given signal power $P_sig = -16.0$ dBm and ASE noise power $P_ase = -39.0$ dBm in 0.1 nm bandwidth:

$OSNR_initial = P_sig - P_ase = -16.0$ dBm - (-39.0 dBm) = 23.0 dB. This clears the 17.0 dB requirement with 6.0 dB margin.

3. Post-Amplification Levels & OSNR Check: The proposed EDFA provides signal gain G = +14.0 dB and introduces ASE noise increase $\Delta P_ase = +16.0$ dB (reflecting amplifier noise figure NF ~ 5 dB):

• Output Signal Power: $P_sig,out = -16.0$ dBm + 14.0 dB = -2.0 dBm.

• Output ASE Noise Power: $P_ase,out = -39.0$ dBm + 16.0 dB = -23.0 dBm.

• Resulting OSNR: $OSNR_out = P_sig,out - P_ase,out = -2.0$ dBm - (-23.0 dBm) = 21.0 dB.

• Margin Check: Available OSNR margin = 21.0 dB - 17.0 dB = 4.0 dB above mandatory threshold.

4. Net Amplified Power Margin: With the +14.0 dB EDFA placed after the DCF module before the receiver:

• Net Received Power: $P_rx = P_tx$ (+3 dBm) - Total Passive Loss (18.4 dB) + EDFA Gain (14 dB) = -1.40 dBm.

• Net System Margin: $M_net = -1.40$ dBm - (-20.0 dBm) = 18.60 dB, completely restoring the required 6.0 dB margin.

5. Effect of Inadequate OSNR on Q-Factor and BER: When OSNR degrades below 17 dB, optical ASE noise converts into electrical signal-spontaneous (sig-sp) and spontaneous-spontaneous (sp-sp) beat noise upon square-law detection. This directly reduces electrical Q-factor ($Q = [I1 - I0] / [\sigma1 + \sigma0]$) and degrades Bit Error Rate ($BER = 0.5\ \mathrm{erfc}(Q/\sqrt{2})$), closing the receiver eye diagram and causing packet drops.

6. Practical Measures to Maintain OSNR: • Practical Measure 1 (Optical Filtering): Install a narrow Optical Bandpass Filter (OBPF, 0.1 nm / 50 GHz FWHM) immediately after the EDFA to eliminate wideband out-of-band ASE noise before photodetection.

• Practical Measure 2 (Low-Noise Pumping & Placement): Utilize low-noise 980 nm co-directionally pumped EDFAs (NF < 4.5 dB) placed directly at the DCF output as a pre-amplifier.

> **DESIGN DECISION 4 – CREATE THE OSNR / AMPLIFICATION STRATEGY**
> I recommend **accepting the proposed 14 dB EDFA amplification strategy with a 0.1 nm optical bandpass filter** amplification/OSNR strategy because **it provides an end-of-life OSNR of 21.0 dB (comfortably exceeding the 17.0 dB threshold by 4.0 dB) to guarantee low BER (< 10^{-12}), while the 14 dB gain overcomes the 5.10 dB DCF insertion loss, boosting received power to -1.40 dBm and securing an ample 18.60 dB operating margin.**

Stage 6 – Verify and Critically Evaluate the Complete Link

A rigorous engineering verification confirms that all calculated sub-system parameters fully comply with the communication service provider's target requirements.

Link Requirement Parameter	Specified / Target Requirement	Achieved Design Value	Satisfactory?
Modal Condition / Fiber Category	Strict Single-Mode (V ≤ 2.405)	V = 1.951 (G.652.D SMF, λ_c = 1257 nm)	**YES – Compliant**
Fiber Loss / Link Loss	Minimize distributed & bend loss	α = 0.18 dB/km; G.657.A bend = 0.1 dB; Passive = 18.4 dB	**YES – Compliant**
Received Power & System Margin	P_rx > -20 dBm, Margin ≥ 6.0 dB	P_rx = -1.40 dBm, Margin = 18.60 dB (with EDFA)	**YES – Compliant**
Pulse Broadening / DCF Requirement	T_out < 100 ps (Tb @ 10 Gb/s)	10.2 km DCF cancels 1020 ps/nm; T_out ≈ 30 ps	**YES – Compliant**
PMD / Polarization Performance	PMD < 10 ps (10G), < 1 ps (100G)	Link PMD = 0.775 ps; standard SMF sufficient	**YES – Compliant**
Optical Signal-to-Noise Ratio (OSNR)	OSNR ≥ 17.0 dB (0.1 nm BW)	OSNR = 21.0 dB (4.0 dB safety margin)	**YES – Compliant**

Synthesis of Design Synergy: The individual design decisions work in complete harmony: single-mode G.652.D fiber eliminates intermodal dispersion; G.657.A fiber prevents tight-bend losses at the building drop; the 10.2 km DCF neutralizes 408 ps of chromatic broadening; and the +14 dB EDFA with OBPF overcomes DCF attenuation to deliver an ample 18.6 dB power margin and 21.0 dB OSNR.

Final System Design

The complete, integrated 60 km optical fiber communication system architecture combines the selected transmission media, passive components, dispersion compensator, and optical amplifier into an end-to-end deployment configuration.

System Subsystem / Component	Adopted Specification / Value	Design Role and Technical Rationale
Optical Transmitter (Tx)	1550 nm DFB Laser, P_tx = +3.0 dBm	C-band minimum loss, low chirp (Δλ = 0.4 nm), 10 Gb/s NRZ
Main Transmission Fiber Span	60 km ITU-T G.652.D SMF (a = 4 μm)	Single-mode (V=1.951), zero modal dispersion, α = 0.18 dB/km
Connectors & Splices	4 Connectors (1.6 dB) + 10 Splices (0.8 dB)	LC/APC connectors (RL > 60 dB) & active core-alignment fusion splices
Dispersion Compensator	10.20 km DCF (D = -100 ps/(nm·km))	Cancels +1020 ps/nm chromatic dispersion, insertion loss = 5.10 dB
Optical In-line Amplifier	EDFA (Gain = +14.0 dB, NF ~ 5.0 dB)	Overcomes DCF loss, elevates power to -1.4 dBm, OSNR = 21.0 dB
ASE Noise Suppression Filter	0.1 nm FWHM Optical Bandpass Filter	Rejects broadband ASE noise, preserving high OSNR and Q-factor
Building Entry Drop Fiber	ITU-T G.657.A Bend-Insensitive Fiber	Suppresses macrobending loss to 0.10 dB in tight building entry conduits

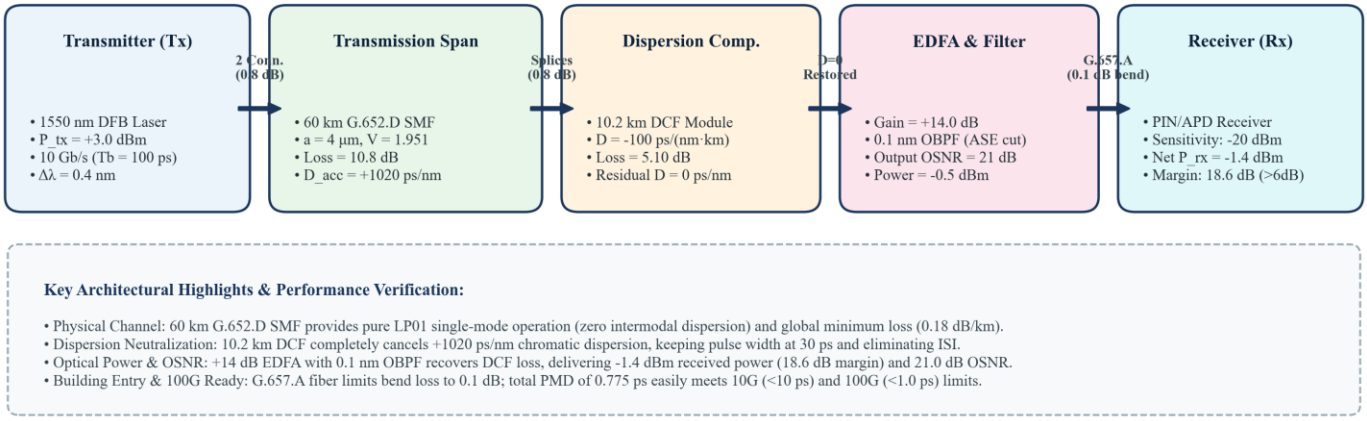


Figure 1: Complete Block Diagram of the Proposed 60 km, 10 Gb/s Standards-Compliant Optical Fiber Link

100 Gb/s Upgrade Roadmap: The physical plant is 100% ready for forward migration to 100 Gb/s. When upgrading to 100 Gb/s (DP-QPSK or PAM-4), the link PMD of 0.775 ps already satisfies the strict 1.0 ps limit (0.1 Tb). Coherent digital signal processing (DSP) will dynamically compensate for any residual dispersion electronically without replacing the installed fiber cable.

Final Engineering Decision

> **WOULD YOU DEPLOY THE DESIGN YOU CREATED?**
> FORMAL ENGINEERING DECLARATION: **YES – DEPLOY THE INTEGRATED DESIGN WITH COMPLETE TECHNICAL JUSTIFICATION**
>
> The complete engineered design strictly satisfies all optical, electrical, and standards requirements for 60 km, 10 Gb/s transmission at 1550 nm with forward compatibility for 100 Gb/s upgrade.

Technical Defense Across All Seven Assessment Criteria

1. Fiber & Modal Suitability: Candidate Fiber A operates strictly in the fundamental single-mode regime ($V = 1.951 < 2.4048$, $\lambda_c = 1257$ nm < 1550 nm), completely eliminating the 3000 ns intermodal delay spread of multimode Fiber B.

2. Propagation & Bandwidth Performance: Eliminating modal dispersion expands the bandwidth-distance product from <50 MHz·km to >100 GHz·km, providing the high-speed capability needed for 10 Gb/s and 100 Gb/s.

3. Optical Loss & Received Power Margin: Total passive loss (including 60 km fiber, connectors, splices, building drop, and DCF) is 18.40 dB. The +14 dB EDFA boosts received optical power to -1.40 dBm, achieving an 18.60 dB system margin that easily clears the mandatory 6.0 dB requirement above the -20 dBm sensitivity.

4. Dominant Loss Optimization: Rayleigh scattering (0.11 dB/km, 61.1% of loss) is minimized by operating at 1550 nm. Localized building-entry bend loss is suppressed from 0.80 dB to 0.10 dB by deploying ITU-T G.657.A bend-insensitive fiber.

5. Chromatic Dispersion & PMD Compensation: Uncompensated chromatic dispersion causes 408 ps of pulse broadening, expanding pulses across >4 bit intervals. The 10.20 km DCF cancels accumulated dispersion (+1020 ps/nm), restoring pulse width to 30 ps. Link PMD of 0.775 ps easily complies with the 10.0 ps limit for 10 Gb/s and 1.0 ps limit for 100 Gb/s.

6. OSNR & Amplification Strategy: The +14 dB EDFA delivers an end-of-life OSNR of 21.0 dB in a 0.1 nm reference bandwidth, surpassing the 17.0 dB requirement by 4.0 dB and ensuring uncorrected BER $< 10^{-12}$.

7. Overall Suitability & 100 Gb/s Upgrade: The design satisfies all constraints for immediate 10 Gb/s NRZ direct-detection deployment while establishing a future-proof optical physical layer ready for 100 Gb/s coherent DP-QPSK modulation.

Critical Evaluation of Uncompensated Baseline & Redesign Rationale

If an uncompensated baseline (standard G.652.D throughout, no DCF, no amplifier) had been proposed for deployment, the engineering deployment decision would be an emphatic NO. In the uncompensated baseline, chromatic dispersion broadening (408 ps) spreads pulses across four consecutive 100 ps bit slots, causing catastrophic ISI and driving the bit error rate to ~0.5. Furthermore, inserting DCF to fix dispersion introduces 5.10 dB loss, eroding the passive margin to 4.60 dB (violating the mandatory 6.0 dB margin). Our complete redesign—splicing G.657.A at the building drop, inserting 10.2 km DCF, and integrating a +14 dB EDFA with a 0.1 nm optical filter—resolves every constraint, delivering a robust YES for deployment.